

IMPE $\text{\LaTeX}$ System

Canonical Multilingual and Feature Showcase

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v1.0.0

Abstract

This document is the canonical visual specimen for the current IMPE $\text{\LaTeX}$ System public interface. It combines a local-family style audit, a deliberately small Unicode-range routing regression, dense multilingual paragraph composition with explicit local overrides, right-to-left text, vertical writing, generic complex shaping, CJK and Han family selection, and the principal document features provided by IMPE. It is intended both as a public showcase and as a release-time visual regression specimen.¹

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¹The showcase is intentionally broader than the small regression examples under `examples/`; those examples remain useful for isolating individual failures.

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1 Local Font Families and Styles

This section presents IMPE’s explicit local font commands and compares their available serif, bold, italic, and sans-serif styles. The complete font-family matrix appears on the following landscape page. Where a requested face is unavailable, IMPE falls back according to the family configuration rather than requiring special-case markup.

2 Multilingual Routing and Mixed-Script Composition

Multilingual composition (combined routing and local overrides) is tested here in two deliberately different ways. The main specimen is a dense paragraph in which explicit local families, right-to-left runs, complex shaping, shared Han choices, and inline vertical writing are allowed to interact. A later compact regression enables automatic Unicode-range ownership for only a small set of ordinary scripts.

2.1 Mixed-Script Paragraph

The following specimen is intentionally one $\text{\TeX}$ paragraph. It contains no English transition sentences: every script that appears contributes a complete sentence or clause. Armenian, Georgian, Coptic, Glagolitic, and Tamil are left as raw Unicode to exercise the small automatic-routing set; all other script families in the paragraph use explicit local overrides.

Հայերեն այս ամբողջական նախադասությունը գրված է նույն բազմալեզու պարբերության մեջ։ յետևյալ ստորագրված լինելով միջազգայնորեն ճանաչված կոդերով և համապատասխանաբար ընթերցվող տեքստերով։ Օրինակ՝ ՄԱԿ-ի ՀԻՄՆԱԿԱԾ ԱՊՐԱՅԻՆ, ԽՍՀՄ-ի ՀԻՄՆԱԿԱԾ ԿՈՄԻՏԵՑԸ։ Սակայն, եթե դուք ունեւում եք համապատասխանաբար ընթերցվող տեքստերով, ապա դուք կունեւում եք համապատասխանաբար ընթերցվող տեքստերով։ Օրինակ՝ ՄԱԿ-ի ՀԻՄՆԱԿԱԾ ԱՊՐԱՅԻՆ, ԽՍՀՄ-ի ՀԻՄՆԱԿԱԾ ԿՈՄԻՏԵՑԸ։

सर्वे भवन्तु सुखिनः, सर्वे सन्तु निरामयाः। தமிழில் எழுதப்பட்ட இந்த முழு வாக்கியம் அதே பலமொழிப் பத்திக்குள் தொடர்ந்து வருகிறது. རྟོག་པོ་རྣམས་ཀྱི་ཚ་བ་ཡིན། هذه جملة عربية كاملة تظهر في وسط الفقرة نفسها وتختبر الانتقال إلى

یہ ایک مکمل اردو جملہ ہے جو اسی کثیر لسانی پیرا گراف کے اندر نستعلیق رسم الخط کی کتابت سے الیمین الی الیسار۔ זהו משפט עברי שלם שממשיך באותה פסקה לאחר הטקסט הערבי והאורדו. میں لکھی گئی ہے۔

日本語のこの文章では、ひらがなとカタカナと漢字が同じ文の中に自然に現れます。한국어의 이 문장에서는 한글과 한자가 같은 문장 안에 함께 나타납니다。這個中文完整句子使用明確指定的繁體中文字體，並與

Family	Command	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
Latin and general-purpose families									
Noto	\NOT	Abg fi fl	Abg fi fl	<i>Abg fi fl</i>	<i>Abg fi fl</i>	Abg fi fl	Abg fi fl	<i>Abg fi fl</i>	<i>Abg fi fl</i>
Times	\TIM	Abg fi fl	Abg fi fl	<i>Abg fi fl</i>	<i>Abg fi fl</i>	Abg fi fl	Abg fi fl	<i>Abg fi fl</i>	<i>Abg fi fl</i>
Gentium Plus	\GEN	Ārya ṛ ṁ ḥ	Ārya ṛ ṁ ḥ	<i>Ārya ṛ ṁ ḥ</i>	<i>Ārya ṛ ṁ ḥ</i>	Ārya ṛ ṁ ḥ	Ārya ṛ ṁ ḥ	<i>Ārya ṛ ṁ ḥ</i>	<i>Ārya ṛ ṁ ḥ</i>
Charis SIL	\CHA	Ārya ṛ ṁ ḥ	Ārya ṛ ṁ ḥ	<i>Ārya ṛ ṁ ḥ</i>	<i>Ārya ṛ ṁ ḥ</i>	Ārya ṛ ṁ ḥ	Ārya ṛ ṁ ḥ	<i>Ārya ṛ ṁ ḥ</i>	<i>Ārya ṛ ṁ ḥ</i>
Libertinus	\LIB	Abg fi fl	Abg fi fl	<i>Abg fi fl</i>	<i>Abg fi fl</i>	Abg fi fl	Abg fi fl	<i>Abg fi fl</i>	<i>Abg fi fl</i>
European and ancient alphabetic scripts									
Carian	\CA	ΑΒΝ	ΑΒΝ	ΑΒΝ	ΑΒΝ	ΑΒΝ	ΑΒΝ	ΑΒΝ	ΑΒΝ
Coptic	\CO	Ⲛⲟⲩⲧⲉ	Ⲛⲟⲩⲧⲉ	Ⲛⲟⲩⲧⲉ	Ⲛⲟⲩⲧⲉ	Ⲛⲟⲩⲧⲉ	Ⲛⲟⲩⲧⲉ	Ⲛⲟⲩⲧⲉ	Ⲛⲟⲩⲧⲉ
Bopomofo	\ZY	ㄗ ㄨ ˊ ㄣ	ㄗ ㄨ ˊ ㄣ	ㄗ ㄨ ˊ ㄣ	ㄗ ㄨ ˊ ㄣ	ㄗ ㄨ ˊ ㄣ	ㄗ ㄨ ˊ ㄣ	ㄗ ㄨ ˊ ㄣ	ㄗ ㄨ ˊ ㄣ
Cuneiform	\CU	𐎶𐎵𐎶𐎶	𐎶𐎵𐎶𐎶	𐎶𐎵𐎶𐎶	𐎶𐎵𐎶𐎶	𐎶𐎵𐎶𐎶	𐎶𐎵𐎶𐎶	𐎶𐎵𐎶𐎶	𐎶𐎵𐎶𐎶
Glagolitic	\GL	ⱭⱭⱭⱭ	ⱭⱭⱭⱭ	ⱭⱭⱭⱭ	ⱭⱭⱭⱭ	ⱭⱭⱭⱭ	ⱭⱭⱭⱭ	ⱭⱭⱭⱭ	ⱭⱭⱭⱭ
Old Italic	\OI	𐌲𐌱𐌴	𐌲𐌱𐌴	𐌲𐌱𐌴	𐌲𐌱𐌴	𐌲𐌱𐌴	𐌲𐌱𐌴	𐌲𐌱𐌴	𐌲𐌱𐌴
Old Hungarian	\OH	ꞤꞤꞤ	ꞤꞤꞤ	ꞤꞤꞤ	ꞤꞤꞤ	ꞤꞤꞤ	ꞤꞤꞤ	ꞤꞤꞤ	ꞤꞤꞤ
Runic	\RU	ᚱᚱᚱ	ᚱᚱᚱ	ᚱᚱᚱ	ᚱᚱᚱ	ᚱᚱᚱ	ᚱᚱᚱ	ᚱᚱᚱ	ᚱᚱᚱ
South and Central Asian scripts									
Armenian	\HY	Հայերեն	Հայերեն	Հայերեն	Հայերեն	Հայերեն	Հայերեն	Հայերեն	Հայերեն
Hindi	\HI	हिन्दी	हिन्दी	हिन्दी	हिन्दी	हिन्दी	हिन्दी	हिन्दी	हिन्दी
Sanskrit	\SA	संस्कृतम्	संस्कृतम्	संस्कृतम्	संस्कृतम्	संस्कृतम्	संस्कृतम्	संस्कृतम्	संस्कृतम्
Devanagari	\DEV	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी
Tamil	\TA	தமிழ்	தமிழ்	தமிழ்	தமிழ்	தமிழ்	தமிழ்	தமிழ்	தமிழ்
Brahmi	\BR	𑀧𑁆𑀭𑀸𑀓	𑀧𑁆𑀭𑀸𑀓	𑀧𑁆𑀭𑀸𑀓	𑀧𑁆𑀭𑀸𑀓	𑀧𑁆𑀭𑀸𑀓	𑀧𑁆𑀭𑀸𑀓	𑀧𑁆𑀭𑀸𑀓	𑀧𑁆𑀭𑀸𑀓
Georgian	\KA	ქართული	ქართული	ქართული	ქართული	ქართული	ქართული	ქართული	ქართული

[illegible]

Family	Command	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
WenJin	\WJ	中國𠂇𠂇	中國𠂇𠂇	中國𠂇𠂇	中國𠂇𠂇	中國𠂇𠂇	中國𠂇𠂇	中國𠂇𠂇	中國𠂇𠂇
Korean	\KR	한국어漢字	한국어漢字	한국어漢字	한국어漢字	한국어漢字	한국어漢字	한국어漢字	한국어漢字
Tangut	\TG	𐞪𐞩𐞰𐞳	𐞪𐞩𐞰𐞳	𐞪𐞩𐞰𐞳	𐞪𐞩𐞰𐞳	𐞪𐞩𐞰𐞳	𐞪𐞩𐞰𐞳	𐞪𐞩𐞰𐞳	𐞪𐞩𐞰𐞳
Mongolian-family and other Asian scripts									
Khitan Small Script (linear)	\KHS	𐰆𐰪𐰏𐰤	𐰆𐰪𐰏𐰤	𐰆𐰪𐰏𐰤	𐰆𐰪𐰏𐰤	𐰆𐰪𐰏𐰤	𐰆𐰪𐰏𐰤	𐰆𐰪𐰏𐰤	𐰆𐰪𐰏𐰤
Mongolian	\MO	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ
Mongolian Baiti	\MOb	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ	ᠮᠣᠩᠭᠣᠯᠠᠨ
Manchu	\MC	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ
Segoe Historic	\SEG	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ	ᡤᡠᡨᡠᡳ
Thai	\TH	ภาษาไทย	ภาษาไทย	ภาษาไทย	ภาษาไทย	ภาษาไทย	ภาษาไทย	ภาษาไทย	ภาษาไทย
Old Turkic	\OT	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏
Old Uyghur	\UY	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏	𐰇𐰆𐰪𐰏
Vietnamese									
Quốc Ngữ	\VI	Tiếng Việt	Tiếng Việt	Tiếng Việt	Tiếng Việt	Tiếng Việt	Tiếng Việt	Tiếng Việt	Tiếng Việt
Hán-Nôm	\HN	𐤙𐤓𐤕𐤔𐤕	𐤙𐤓𐤕𐤔𐤕	𐤙𐤓𐤕𐤔𐤕	𐤙𐤓𐤕𐤔𐤕	𐤙𐤓𐤕𐤔𐤕	𐤙𐤓𐤕𐤔𐤕	𐤙𐤓𐤕𐤔𐤕	𐤙𐤓𐤕𐤔𐤕

前後不同文種保持在同一個段落中。

의 몽골 문자의 직후에도日本語の完全な文章が続き、通常の横書き状態が回復したことを確認します。ᠮᠤᠩᠭᠣᠯᠡᠨᠪᠢᠴᠦ᠋ᠳᠤᠨᠵᠢᠰᠡᠶ᠋ᠠᠨᠠᠷᠠᠭ᠎ᠠᠵᠢᠸᠠᠭᠤᠨᠲᠤᠨᠠᠷᠠᠭᠤᠨᠠᠨᠠᠵᠢᠸᠠᠭᠤᠨ두 번째 세로쓰기 구간 뒤에도 한국어의 완전한 문장이 이어지며 문단의 방향과 글꼴 상태가 정상적으로 복원되는지를 확인합니다. 最後這個中文完整句子仍然留在同一個段落裡，用來確認兩次豎排 local override、數次 RTL 切換與複雜 shaping 之後，普通橫排行文與局部字體狀態全部恢復正常。

2.2 Minimal Automatic Unicode-Range Regression

Unicode-range routing (automatic font ownership) is exercised here only on a small representative set. This keeps the v1.0.0 showcase focused on routing correctness without turning the entire document into a large interchar transition graph. The specimen itself contains raw Unicode only.

[illegible]

3 Right-to-Left Scripts

Right-to-left text (directional behavior) requires more than choosing a glyph source. This section therefore uses explicit local families throughout. Each specimen is long enough to form a genuine paragraph, allowing line breaking, inter-word spacing, punctuation, digits, bidirectional interaction, and OpenType shaping to be inspected without enabling overlapping automatic Arabic, Hebrew, or Syriac range owners.

3.1 Arabic

الْعَرَبِيَّةُ لَفَةً ذَاتُ تَارِيخٍ طَوِيلٍ، وَهِيَ تُكْتَبُ مِنَ الْيَمِينِ إِلَى الْيَسَارِ، وَتَتَّصِلُ حُرُوفُهَا بِأَشْكَالٍ مُخْتَلِفَةٍ بِحَسَبِ مَوْضِعِهَا فِي الْكَلِمَةِ. يَحْتَوِي هَذَا النَّصُّ عَلَى جُمَلٍ طَوِيلَةٍ وَعَلَامَاتٍ تَرْقِيمٍ مُتَنَوِّعَةٍ، وَتَتَغَيَّرُ أَطْوَالُ الْكَلِمَاتِ وَالْجُمَلِ حَتَّى يَظْهَرَ سُلُوكُ الْفَقْرَةِ عِنْدَ نِهَايَةِ السَّطْرِ. وَعِنْدَمَا يَطُولُ السَّطْرُ يَجِبُ أَنْ يَنْتَقِلَ النَّصُّ إِلَى السَّطْرِ التَّالِي بِصُورَةٍ طَبِيعِيَّةٍ مِنْ غَيْرِ أَنْ تَنْقَطِعَ أَشْكَالُ الْحُرُوفِ أَوْ تَضْطَرِّبَ الْمَسَافَاتُ بَيْنَ الْكَلِمَاتِ. وَبُسْتَعْمَلُ هَذَا الْمُفْطَعِ لِلتَّأَكُّدِ مِنْ

Inline Arabic inside an English sentence: الترقيم في سطر واحد مع علامات الترقيم before returning to ordinary Latin text.

اُردو ایک ایسی زبان ہے جس کی تحریری روایت میں نستعلیق رسم الخط کو خاص اہمیت حاصل ہے۔ اس رسم الخط میں حروف صرف ایک دوسرے سے ملتے ہی نہیں بلکہ الفاظ کی مجموعی ساخت بھی سطر کی بصری صورت پر اثر ڈالتی ہے، اس لیے طویل عبارت مختصر نمونے سے کہیں زیادہ مفید آزمائش فراہم کرتی ہے۔ اس پیراگراف میں مختلف لمبائی کے الفاظ، مختلف رموزِ اوقاف، اور کئی مسلسل جملے شامل ہیں تاکہ دائیں سے بائیں بہاؤ، لفظی فاصلوں، اور سطر کے اختتام پر عبارت کی تقسیم کو واضح طور پر دیکھا جاسکے۔ جب عبارت ایک سطر میں پوری نہ آئے تو اگلی سطر میں منتقلی قدرتی ہونی چاہیے، الفاظ کی ترتیب برقرار رہنی چاہیے، اور نستعلیق کی تشکیل کسی غلط فونٹ یا غیر متعلقہ عربی طرز میں تبدیل نہیں ہونی چاہیے۔ یہ آخری جملہ خاص طور پر اس بات کی جانچ کرتا ہے کہ ایک مکمل اُردو پیراگراف کئی سطروں میں تقسیم ہونے کے بعد بھی اپنی مقامی نستعلیق فیملی اور درست سمت برقرار رکھتا ہے۔

העברית נכתבת מימין לשמאל, ולכן פסקה ארוכה מאפשרת לבדוק לא רק את צורת האותיות אלא גם את התנהגות השורות, הרווחים וסימני הפיסוק. בטקסט הזה מופיעים משפטים באורכים שונים כדי לבדוק כיצד המערכת מתמודדת עם רצף עברי ארוך בתוך פסקה רגילה. כאשר השורה נעשית ארוכה מדי, המעבר לשורה הבאה צריך להתרחש במקום טבעי בלי לשנות את סדר המילים ובלי לפגוע במיקום הנקודות, הפסיקים והסוגריים. אפשר גם לכתוב מילים מנוקדות כמו שָׁלוֹם, עֵבְרִית וְכִתְיָבָה בתוך אותה פסקה, וכך לבדוק שהניקוד נשאר מחובר לאות הנכונה גם לאחר שבירת שורה. המשפט האחרון נועד לוודא שהפסקה כולה נשארת יציבה בתוך מסמך רב־לשוני ולאחריה חוזר הטקסט הסובב להתנהגות הרגילה.

על פי חוק, כל מי שיש לו זכויות בעלות על נדלן, חייב לשלם מס על הנדלן. מס על הנדלן הוא מס שנתי, שמשלם אותו הבעלים של הנדלן. מס על הנדלן הוא מס שנתי, שמשלם אותו הבעלים של הנדלן.

Eastern Syriac stylistic family:

[illegible]

Vertical writing (horizontal-layout-compatible vertical embedding) is currently implemented as a compatibility mode inside an otherwise horizontal document layout. IMPE does not yet provide a fully vertical page or paragraph flow. Instead, the text is divided at word boundaries, and each word is typeset vertically while the resulting word-sized units continue to participate in ordinary horizontal paragraph layout. Spaces therefore become natural break opportunities between vertical words.

This design keeps genuinely vertical glyph orientation while allowing the text to wrap within normal horizontal pages, with relatively compact line spacing and without requiring a separate vertical document model. The specimens below are deliberately long enough to test this compatibility strategy under realistic paragraph conditions: word segmentation, inter-word spacing, line breaking, line spacing, punctuation, ordering, and recovery of the surrounding horizontal layout.

Horizontal

[illegible]

١. ٢٠٠٠	٢. ٢٠٠٠	٣. ٢٠٠٠	٤. ٢٠٠٠	٥. ٢٠٠٠	٦. ٢٠٠٠	٧. ٢٠٠٠	٨. ٢٠٠٠	٩. ٢٠٠٠	١٠. ٢٠٠٠
١١. ٢٠٠٠	١٢. ٢٠٠٠	١٣. ٢٠٠٠	١٤. ٢٠٠٠	١٥. ٢٠٠٠	١٦. ٢٠٠٠	١٧. ٢٠٠٠	١٨. ٢٠٠٠	١٩. ٢٠٠٠	٢٠. ٢٠٠٠
٢١. ٢٠٠٠	٢٢. ٢٠٠٠	٢٣. ٢٠٠٠	٢٤. ٢٠٠٠	٢٥. ٢٠٠٠	٢٦. ٢٠٠٠	٢٧. ٢٠٠٠	٢٨. ٢٠٠٠	٢٩. ٢٠٠٠	٣٠. ٢٠٠٠
٣١. ٢٠٠٠	٣٢. ٢٠٠٠	٣٣. ٢٠٠٠	٣٤. ٢٠٠٠	٣٥. ٢٠٠٠	٣٦. ٢٠٠٠	٣٧. ٢٠٠٠	٣٨. ٢٠٠٠	٣٩. ٢٠٠٠	٤٠. ٢٠٠٠
٤١. ٢٠٠٠	٤٢. ٢٠٠٠	٤٣. ٢٠٠٠	٤٤. ٢٠٠٠	٤٥. ٢٠٠٠	٤٦. ٢٠٠٠	٤٧. ٢٠٠٠	٤٨. ٢٠٠٠	٤٩. ٢٠٠٠	٥٠. ٢٠٠٠
٥١. ٢٠٠٠	٥٢. ٢٠٠٠	٥٣. ٢٠٠٠	٥٤. ٢٠٠٠	٥٥. ٢٠٠٠	٥٦. ٢٠٠٠	٥٧. ٢٠٠٠	٥٨. ٢٠٠٠	٥٩. ٢٠٠٠	٦٠. ٢٠٠٠
٦١. ٢٠٠٠	٦٢. ٢٠٠٠	٦٣. ٢٠٠٠	٦٤. ٢٠٠٠	٦٥. ٢٠٠٠	٦٦. ٢٠٠٠	٦٧. ٢٠٠٠	٦٨. ٢٠٠٠	٦٩. ٢٠٠٠	٧٠. ٢٠٠٠
٧١. ٢٠٠٠	٧٢. ٢٠٠٠	٧٣. ٢٠٠٠	٧٤. ٢٠٠٠	٧٥. ٢٠٠٠	٧٦. ٢٠٠٠	٧٧. ٢٠٠٠	٧٨. ٢٠٠٠	٧٩. ٢٠٠٠	٨٠. ٢٠٠٠
٨١. ٢٠٠٠	٨٢. ٢٠٠٠	٨٣. ٢٠٠٠	٨٤. ٢٠٠٠	٨٥. ٢٠٠٠	٨٦. ٢٠٠٠	٨٧. ٢٠٠٠	٨٨. ٢٠٠٠	٨٩. ٢٠٠٠	٩٠. ٢٠٠٠
٩١. ٢٠٠٠	٩٢. ٢٠٠٠	٩٣. ٢٠٠٠	٩٤. ٢٠٠٠	٩٥. ٢٠٠٠	٩٦. ٢٠٠٠	٩٧. ٢٠٠٠	٩٨. ٢٠٠٠	٩٩. ٢٠٠٠	١٠٠. ٢٠٠٠

4.2 Manchu

Horizontal

ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ
ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ
ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ
ᡤᡠᡵᡠ ᡤᡠᡵᡠ

Vertical

ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ
ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ
ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ

4.3 Old Uyghur

The Old Uyghur sample is deliberately extended so that several successive vertical columns must be produced. It is used as a layout stress string rather than as a continuous philological quotation.

Horizontal

ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ
ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ
ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ

Vertical

ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ
ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ ᡤᡠᡵᡠ

5 Complex Shaping and Stacked Sequences

OpenType shaping (generic complex shaping) is tested here through explicit local families. This deliberately separates shaping from Unicode-range ownership: if a local specimen is wrong, the failure belongs to the selected family or shaping path rather than to an automatic range transition. Egyptian Hieroglyphs remain outside this showcase, and Khitan Small Script is recorded only in its current linear fallback form.

5.1 Devanagari

The Sanskrit local family exercises conjuncts, virama behavior, pre-base and post-base vowel signs, and longer clusters without enabling a document-wide Devanagari owner.

A longer local specimen:

धर्मक्षेत्रे कुरुक्षेत्रे समवेता युयुत्सवः । संस्कृतवाक्यानि देवनागर्या लिखितानि सन्ति ।

Table 2: Devanagari shaping specimens

Sequence	Specimen
kṣa	क्ष
jña	ज्ञ
śra	श्र
stra	स्त्र
ddhya	द्ध्य
dharma	धर्म
dharmakṣetre	धर्मक्षेत्रे
saṃskṛtam	संस्कृतम्
om	ॐ

5.2 Tibetan

Tibetan provides a visible example of generic vertical stacking inside one local shaping run.

Table 3: Tibetan stack specimens

Type	Specimen
stack	བསྐྱབས་
stack	བརྒྱད་
stack	བཟླ་
stack	རྒྱལ་
stack	དབྱངས་

A longer local paragraph exercises Tibetan shaping and line breaking under ordinary document width:

བོད་ཡིག་ནི་བོད་ཀྱི་སྐད་ཡིག་འབྲི་བའི་ཡི་གེའི་མ་ལག་ཅིག་ཡིན་ཞིང་། ཆོག་དང་ཆོག་གི་བར་ལ་ཆོག་གིས་
མཚམས་འབྲེད་པ་དང་། ཡི་གེ་མང་པོ་ཞིག་མགོ་ཅན་དང་འདོགས་ཅན་དུ་སྒྱར་ནས་ཆོག་བར་གཅིག་གི་ནང་དུ་སྦྱངས་ཏེ་
འབྲི་དགོས། དཔེར་ན་བསྐྱབས་པ་དང་། བརྒྱད་པ། བཟླ་པ། རྒྱལ་ཁབ། དབྱངས་ཅན་ལྟ་བུའི་ཆོག་ནང་དུ་ཡི་གེ་མང་པོ་
སྒྱར་ཡོད་པས། ཡིག་གཟུགས་ཀྱི་སྦྱངས་སྦྲངས་དང་ཆོག་བར་གྱི་བར་ཐག་གཉིས་ཀ་ཡང་དག་པར་སྟོན་དགོས། ཆོག་ཕྱིང་
རིང་པོ་ཞིག་ཤོག་ངོས་ཀྱི་ཞིང་ཆད་ལས་བཀལ་བའི་སྐབས་སུ། ཆོག་གི་རྗེས་སུ་རང་བཞིན་གྱིས་ཕྱིང་བ་གསར་པར་འགོ་
ཐུབ་པ་དང་། སྔ་ཕྱིའི་ཡི་གེའི་སྦྱངས་སྦྲངས་དང་ཆོག་གི་གོ་རིམ་མི་འབྲུག་པ་གལ་ཆེ། དེའི་ཕྱིར་ཆོག་ཕྱིང་རིང་པོ་འདིས་
བོད་ཡིག་གི་གཟུགས་སྟོར་དང་། ཆོག་བར། ཕྱིང་བར། ཤད་དང་ཉིས་ཤད་ཀྱི་གནས་སྦྲངས། དེ་བཞིན་དུ་ཕྱིང་བ་བརྗེ་བའི་
སྐབས་ཀྱི་སྤྱིའི་སྒྲིག་སྦྲངས་བཅས་ལ་ཆོད་ལྟ་བུ་དུ་སྦྱོར།

5.3 Brahmi, Kharoshthi, and Thai

Brahmi is provided through two local families. The redistributable Noto family provides broad Brahmi coverage, but in the current IMPE/X_YTeX setup it does not produce the expected stacked forms for all conjunct sequences. Segoe UI Historic is therefore exposed through `\SEG` as an alternative reference family and produces the intended stacking in these cases. Because Segoe UI Historic is a Microsoft Windows font, its font binary is not distributed with IMPE; the local command is available only when the user supplies or already has access to the font.

Table 4: Additional generic-shaping specimens

Script / family	Specimen
Brahmi / Noto	𑀀𑀁𑀂 𑀃𑀄, Dīṣṣ 𑀅𑀆.
Brahmi / Segoe Historic	𑀇𑀈 𑀉𑀊, Dīṣṣ 𑀋𑀌.
Kharoshthi	𑀀𑀁𑀂𑀃𑀄𑀅𑀆𑀇𑀈𑀉𑀊𑀋𑀌𑀍𑀎𑀏𑀐𑀑𑀒𑀓𑀔𑀕𑀖𑀗𑀘𑀙𑀚𑀛𑀜𑀝𑀞𑀟𑀠𑀡𑀢𑀣𑀤𑀥𑀦𑀧𑀨𑀩𑀪𑀫𑀬𑀭𑀮𑀯𑀰𑀱𑀲𑀳𑀴𑀵𑀶𑀷𑀸𑀹𑀺𑀻𑀼𑀽𑀾𑀿𑁀𑁁𑁂𑁃𑁄𑁅𑁆𑁇𑁈𑁉𑁊𑁋𑁌𑁍𑁎𑁏𑁐𑁑𑁒𑁓𑁔𑁕𑁖𑁗𑁘𑁙𑁚𑁛𑁜𑁝𑁞𑁟𑁠𑁡𑁢𑁣𑁤𑁥𑁦𑁧𑁨𑁩𑁪𑁫𑁬𑁭𑁮𑁯𑁰𑁱𑁲𑁳𑁴𑁵𑁶𑁷𑁸𑁹𑁺𑁻𑁼𑁽𑁾𑁿𑂀𑂁𑂂𑂃𑂄𑂅𑂆𑂇𑂈𑂉𑂊𑂋𑂌𑂍𑂎𑂏𑂐𑂑𑂒𑂓𑂔𑂕𑂖𑂗𑂘𑂙𑂚𑂛𑂜𑂝𑂞𑂟𑂠𑂡𑂢𑂣𑂤𑂥𑂦𑂧𑂨𑂩𑂪𑂫𑂬𑂭𑂮𑂯𑂰𑂱𑂲𑂳𑂴𑂵𑂶𑂷𑂸𑂺𑂹𑂻𑂼𑂽𑂾𑂿𑃀𑃁𑃂𑃃𑃄𑃅𑃆𑃇𑃈𑃉𑃊𑃋𑃌𑃍𑃎𑃏𑃐𑃑𑃒𑃓𑃔𑃕𑃖𑃗𑃘𑃙𑃚𑃛𑃜𑃝𑃞𑃟𑃠𑃡𑃢𑃣𑃤𑃥𑃦𑃧𑃨𑃩𑃪𑃫𑃬𑃭𑃮𑃯𑃰𑃱𑃲𑃳𑃴𑃵𑃶𑃷𑃸𑃹𑃺𑃻𑃼𑃽𑃾𑃿𑄀𑄁𑄂𑄃𑄄𑄅𑄆𑄇𑄈𑄉𑄊𑄋𑄌𑄍𑄎𑄏𑄐𑄑𑄒𑄓𑄔𑄕𑄖𑄗𑄘𑄙𑄚𑄛𑄜𑄝𑄞𑄟𑄠𑄡𑄢𑄣𑄤𑄥𑄦𑄧𑄨𑄩𑄪𑄫𑄬𑄭𑄮𑄯𑄰𑄱𑄲𑄳𑄴𑄵𑄶𑄷𑄸𑄹𑄺𑄻𑄼𑄽𑄾𑄿𑅀𑅁𑅂𑅃𑅄𑅅𑅆𑅇𑅈𑅉𑅊𑅋𑅌𑅍𑅎𑅏𑅐𑅑𑅒𑅓𑅔𑅕𑅖𑅗𑅘𑅙𑅚𑅛𑅜𑅝𑅞𑅟𑅠𑅡𑅢𑅣𑅤𑅥𑅦𑅧𑅨𑅩𑅪𑅫𑅬𑅭𑅮𑅯𑅰𑅱𑅲𑅳𑅴𑅵𑅶𑅷𑅸𑅹𑅺𑅻𑅼𑅽𑅾𑅿𑆀𑆁𑆂𑆃𑆄𑆅𑆆𑆇𑆈𑆉𑆊𑆋𑆌𑆍𑆎𑆏𑆐𑆑𑆒𑆓𑆔𑆕𑆖𑆗𑆘𑆙𑆚𑆛𑆜𑆝𑆞𑆟𑆠𑆡𑆢𑆣𑆤𑆥𑆦𑆧𑆨𑆩𑆪𑆫𑆬𑆭𑆮𑆯𑆰𑆱𑆲𑆳𑆴𑆵𑆶𑆷𑆸𑆹𑆺𑆻𑆼𑆽𑆾𑆿𑇀𑇁𑇂𑇃𑇄𑇅𑇆𑇇𑇈𑇉𑇊𑇋𑇌𑇍𑇎𑇏𑇐𑇑𑇒𑇓𑇔𑇕𑇖𑇗𑇘𑇙𑇚𑇛𑇜𑇝𑇞𑇟𑇠𑇡𑇢𑇣𑇤𑇥𑇦𑇧𑇨𑇩𑇪𑇫𑇬𑇭𑇮𑇯𑇰𑇱𑇲𑇳𑇴𑇵𑇶𑇷𑇸𑇹𑇺𑇻𑇼𑇽𑇾𑇿𑈀𑈁𑈂𑈃𑈄𑈅𑈆𑈇𑈈𑈉𑈊𑈋𑈌𑈍𑈎𑈏𑈐𑈑𑈒𑈓𑈔𑈕𑈖𑈗𑈘𑈙𑈚𑈛𑈜𑈝𑈞𑈟𑈠𑈡𑈢𑈣𑈤𑈥𑈦𑈧𑈨𑈩𑈪𑈫𑈬𑈭𑈮𑈯𑈰𑈱𑈲𑈳𑈴𑈶𑈵𑈷𑈸𑈹𑈺𑈻𑈼𑈽𑈾𑈿𑉀𑉁𑉂𑉃𑉄𑉅𑉆𑉇𑉈𑉉𑉊𑉋𑉌𑉍𑉎𑉏𑉐𑉑𑉒𑉓𑉔𑉕𑉖𑉗𑉘𑉙𑉚𑉛𑉜𑉝𑉞𑉟𑉠𑉡𑉢𑉣𑉤𑉥𑉦𑉧𑉨𑉩𑉪𑉫𑉬𑉭𑉮𑉯𑉰𑉱𑉲𑉳𑉴𑉵𑉶𑉷𑉸𑉹𑉺𑉻𑉼𑉽𑉾𑉿𑊀𑊁𑊂𑊃𑊄𑊅𑊆𑊇𑊈𑊉𑊊𑊋𑊌𑊍𑊎𑊏𑊐𑊑𑊒𑊓𑊔𑊕𑊖𑊗𑊘𑊙𑊚𑊛𑊜𑊝𑊞𑊟𑊠𑊡𑊢𑊣𑊤𑊥𑊦𑊧𑊨𑊩𑊪𑊫𑊬𑊭𑊮𑊯𑊰𑊱𑊲𑊳𑊴𑊵𑊶𑊷𑊸𑊹𑊺𑊻𑊼𑊽𑊾𑊿𑋀𑋁𑋂𑋃𑋄𑋅𑋆𑋇𑋈𑋉𑋊𑋋𑋌𑋍𑋎𑋏𑋐𑋑𑋒𑋓𑋔𑋕𑋖𑋗𑋘𑋙𑋚𑋛𑋜𑋝𑋞𑋟𑋠𑋡𑋢𑋣𑋤𑋥𑋦𑋧𑋨𑋩𑋪𑋫𑋬𑋭𑋮𑋯𑋰𑋱𑋲𑋳𑋴𑋵𑋶𑋷𑋸𑋹𑋺𑋻𑋼𑋽𑋾𑋿𑌀𑌁𑌂𑌃𑌄𑌅𑌆𑌇𑌈𑌉𑌊𑌋𑌌𑌍𑌎𑌏𑌐𑌑𑌒𑌓𑌔𑌕𑌖𑌗𑌘𑌙𑌚𑌛𑌜𑌝𑌞𑌟𑌠𑌡𑌢𑌣𑌤𑌥𑌦𑌧𑌨𑌩𑌪𑌫𑌬𑌭𑌮𑌯𑌰𑌱𑌲𑌳𑌴𑌵𑌶𑌷𑌸𑌹𑌺𑌻𑌼𑌽𑌾𑌿𑍀𑍁𑍂𑍃𑍄𑍅𑍆𑍇𑍈𑍉𑍊𑍋𑍌𑍍𑍎𑍏𑍐𑍑𑍒𑍓𑍔𑍕𑍖𑍗𑍘𑍙𑍚𑍛𑍜𑍝𑍞𑍟𑍠𑍡𑍢𑍣𑍤𑍥𑍦𑍧𑍨𑍩𑍪𑍫𑍬𑍭𑍮𑍯𑍰𑍱𑍲𑍳𑍴𑍵𑍶𑍷𑍸𑍹𑍺𑍻𑍼𑍽𑍾𑍿𑎀𑎁𑎂𑎃𑎄𑎅𑎆𑎇𑎈𑎉𑎊𑎋𑎌𑎍𑎎𑎏𑎐𑎑𑎒𑎓𑎔𑎕𑎖𑎗𑎘𑎙𑎚𑎛𑎜𑎝𑎞𑎟𑎠𑎡𑎢𑎣𑎤𑎥𑎦𑎧𑎨𑎩𑎪𑎫𑎬𑎭𑎮𑎯𑎰𑎱𑎲𑎳𑎴𑎵𑎶𑎷𑎸𑎹𑎺𑎻𑎼𑎽𑎾𑎿𑏀𑏁𑏂𑏃𑏄𑏅𑏆𑏇𑏈𑏉𑏊𑏋𑏌𑏍𑏎𑏏𑏐𑏑𑏒𑏓𑏔𑏕𑏖𑏗𑏘𑏙𑏚𑏛𑏜𑏝𑏞𑏟𑏠𑏡𑏢𑏣𑏤𑏥𑏦𑏧𑏨𑏩𑏪𑏫𑏬𑏭𑏮𑏯𑏰𑏱𑏲𑏳𑏴𑏵𑏶𑏷𑏸𑏹𑏺𑏻𑏼𑏽𑏾𑏿𑐀𑐁𑐂𑐃𑐄𑐅𑐆𑐇𑐈𑐉𑐊𑐋𑐌𑐍𑐎𑐏𑐐𑐑𑐒𑐓𑐔𑐕𑐖𑐗𑐘𑐙𑐚𑐛𑐜𑐝𑐞𑐟𑐠𑐡𑐢𑐣𑐤𑐥𑐦𑐧𑐨𑐩𑐪𑐫𑐬𑐭𑐮𑐯𑐰𑐱𑐲𑐳𑐴𑐵𑐶𑐷𑐸𑐹𑐺𑐻𑐼𑐽𑐾𑐿𑑀𑑁𑑂𑑃𑑄𑑅𑑆𑑇𑑈𑑉𑑊𑑋𑑌𑑍𑑎𑑏𑑐𑑑𑑒𑑓𑑔𑑕𑑖𑑗𑑘𑑙𑑚𑑛𑑜𑑝𑑞𑑟𑑠𑑡𑑢𑑣𑑤𑑥𑑦𑑧𑑨𑑩𑑪𑑫𑑬𑑭𑑮𑑯𑑰𑑱𑑲𑑳𑑴𑑵𑑶𑑷𑑸𑑹𑑺𑑻𑑼𑑽𑑾𑑿𑒀𑒁𑒂𑒃𑒄𑒅𑒆𑒇𑒈𑒉𑒊𑒋𑒌𑒍𑒎𑒏𑒐𑒑𑒒𑒓𑒔𑒕𑒖𑒗𑒘𑒙𑒚𑒛𑒜𑒝𑒞𑒟𑒠𑒡𑒢𑒣𑒤𑒥𑒦𑒧𑒨𑒩𑒪𑒫𑒬𑒭𑒮𑒯𑒰𑒱𑒲𑒳𑒴𑒵𑒶𑒷𑒸𑒻𑒻𑒼𑒽𑒾𑒿𑓀𑓁𑓃𑓂𑓄𑓅𑓆𑓇𑓈𑓉𑓊𑓋𑓌𑓍𑓎𑓏𑓐𑓑𑓒𑓓𑓔𑓕𑓖𑓗𑓘𑓙𑓚𑓛𑓜𑓝𑓞𑓟𑓠𑓡𑓢𑓣𑓤𑓥𑓦𑓧𑓨𑓩𑓪𑓫𑓬𑓭𑓮𑓯𑓰𑓱𑓲𑓳𑓴𑓵𑓶𑓷𑓸𑓹𑓺𑓻𑓼𑓽𑓾𑓿𑔀𑔁𑔂𑔃𑔄𑔅𑔆𑔇𑔈𑔉𑔊𑔋𑔌𑔍𑔎𑔏𑔐𑔑𑔒𑔓𑔔𑔕𑔖𑔗𑔘𑔙𑔚𑔛𑔜𑔝𑔞𑔟𑔠𑔡𑔢𑔣𑔤𑔥𑔦𑔧𑔨𑔩𑔪𑔫𑔬𑔭𑔮𑔯𑔰𑔱𑔲𑔳𑔴𑔵𑔶𑔷𑔸𑔹𑔺𑔻𑔼𑔽𑔾𑔿𑕀𑕁𑕂𑕃𑕄𑕅𑕆𑕇𑕈𑕉𑕊𑕋𑕌𑕍𑕎𑕏𑕐𑕑𑕒𑕓𑕔𑕕𑕖𑕗𑕘𑕙𑕚𑕛𑕜𑕝𑕞𑕟𑕠𑕡𑕢𑕣𑕤𑕥𑕦𑕧𑕨𑕩𑕪𑕫𑕬𑕭𑕮𑕯𑕰𑕱𑕲𑕳𑕴𑕵𑕶𑕷𑕸𑕹𑕺𑕻𑕼𑕽𑕾𑕿𑖀𑖁𑖂𑖃𑖄𑖅𑖆𑖇𑖈𑖉𑖊𑖋𑖌𑖍𑖎𑖏𑖐𑖑𑖒𑖓𑖔𑖕𑖖𑖗𑖘𑖙𑖚𑖛𑖜𑖝𑖞𑖟𑖠𑖡𑖢𑖣𑖤𑖥𑖦𑖧𑖨𑖩𑖪𑖫𑖬𑖭𑖮𑖯𑖰𑖱𑖲𑖳𑖴𑖵𑖶𑖷𑖸𑖹𑖺𑖻𑖼𑖽𑖾𑗀𑖿𑗁𑗂𑗃𑗄𑗅𑗆𑗇𑗈𑗉𑗊𑗋𑗌𑗍𑗎𑗏𑗐𑗑𑗒𑗓𑗔𑗕𑗖𑗗𑗘𑗙𑗚𑗛𑗜𑗝𑗞𑗟𑗠𑗡𑗢𑗣𑗤𑗥𑗦𑗧𑗨𑗩𑗪𑗫𑗬𑗭𑗮𑗯𑗰𑗱𑗲𑗳𑗴𑗵𑗶𑗷𑗸𑗹𑗺𑗻𑗼𑗽𑗾𑗿𑘀𑘁𑘂𑘃𑘄𑘅𑘆𑘇𑘈𑘉𑘊𑘋𑘌𑘍𑘎𑘏𑘐𑘑𑘒𑘓𑘔𑘕𑘖𑘗𑘘𑘙𑘚𑘛𑘜𑘝𑘞𑘟𑘠𑘡𑘢𑘣𑘤𑘥𑘦𑘧𑘨𑘩𑘪𑘫𑘬𑘭𑘮𑘯𑘰𑘱𑘲𑘳𑘴𑘵𑘶𑘷𑘸𑘹𑘺𑘻𑘼𑘽𑘾𑘿𑙀𑙁𑙂𑙃𑙄𑙅𑙆𑙇𑙈𑙉𑙊𑙋𑙌𑙍𑙎𑙏𑙐𑙑𑙒𑙓𑙔𑙕𑙖𑙗𑙘𑙙𑙚𑙛𑙜𑙝𑙞𑙟𑙠𑙡𑙢𑙣𑙤𑙥𑙦𑙧𑙨𑙩𑙪𑙫𑙬𑙭𑙮𑙯𑙰𑙱𑙲𑙳𑙴𑙵𑙶𑙷𑙸𑙹𑙺𑙻𑙼𑙽𑙾𑙿𑚀𑚁𑚂𑚃𑚄𑚅𑚆𑚇𑚈𑚉𑚊𑚋𑚌𑚍𑚎𑚏𑚐𑚑𑚒𑚓𑚔𑚕𑚖𑚗𑚘𑚙𑚚𑚛𑚜𑚝𑚞𑚟𑚠𑚡𑚢𑚣𑚤𑚥𑚦𑚧𑚨𑚩𑚪𑚫𑚬𑚭𑚮𑚯𑚰𑚱𑚲𑚳𑚴𑚵𑚷𑚶𑚸𑚹𑚺𑚻𑚼𑚽𑚾𑚿𑛀𑛁𑛂𑛃𑛄𑛅𑛆𑛇𑛈𑛉𑛊𑛋𑛌𑛍𑛎𑛏𑛐𑛑𑛒𑛓𑛔𑛕𑛖𑛗𑛘𑛙𑛚𑛛𑛜𑛝𑛞𑛟𑛠𑛡𑛢𑛣𑛤𑛥𑛦𑛧𑛨𑛩𑛪𑛫𑛬𑛭𑛮𑛯𑛰𑛱𑛲𑛳𑛴𑛵𑛶𑛷𑛸𑛹𑛺𑛻𑛼𑛽𑛾𑛿𑜀𑜁𑜂𑜃𑜄𑜅𑜆𑜇𑜈𑜉𑜊𑜋𑜌𑜍𑜎𑜏𑜐𑜑𑜒𑜓𑜔𑜕𑜖𑜗𑜘𑜙𑜚𑜛𑜜𑜝𑜞𑜟𑜠𑜡𑜢𑜣𑜤𑜥𑜦𑜧𑜨𑜩𑜪𑜫𑜬𑜭𑜮𑜯𑜰𑜱𑜲𑜳𑜴𑜵𑜶𑜷𑜸𑜹𑜺𑜻𑜼𑜽𑜾𑜿𑝀𑝁𑝂𑝃𑝄𑝅𑝆𑝇𑝈𑝉𑝊𑝋𑝌𑝍𑝎𑝏𑝐𑝑𑝒𑝓𑝔𑝕𑝖𑝗𑝘𑝙𑝚𑝛𑝜𑝝𑝞𑝟𑝠𑝡𑝢𑝣𑝤𑝥𑝦𑝧𑝨𑝩𑝪𑝫𑝬𑝭𑝮𑝯𑝰𑝱𑝲𑝳𑝴𑝵𑝶𑝷𑝸𑝹𑝺𑝻𑝼𑝽𑝾𑝿𑞀𑞁𑞂𑞃𑞄𑞅𑞆𑞇𑞈𑞉𑞊𑞋𑞌𑞍𑞎𑞏𑞐𑞑𑞒𑞓𑞔𑞕𑞖𑞗𑞘𑞙𑞚𑞛𑞜𑞝𑞞𑞟𑞠𑞡𑞢𑞣𑞤𑞥𑞦𑞧𑞨𑞩𑞪𑞫𑞬𑞭𑞮𑞯𑞰𑞱𑞲𑞳𑞴𑞵𑞶𑞷𑞸𑞹𑞺𑞻𑞼𑞽𑞾𑞿𑟀𑟁𑟂𑟃𑟄𑟅𑟆𑟇𑟈𑟉𑟊𑟋𑟌𑟍𑟎𑟏𑟐𑟑𑟒𑟓𑟔𑟕𑟖𑟗𑟘𑟙𑟚𑟛𑟜𑟝𑟞𑟟𑟠𑟡𑟢𑟣𑟤𑟥𑟦𑟧𑟨𑟩𑟪𑟫𑟬𑟭𑟮𑟯𑟰𑟱𑟲𑟳𑟴𑟵𑟶𑟷𑟸𑟹𑟺𑟻𑟼𑟽𑟾𑟿𑠀𑠁𑠂𑠃𑠄𑠅𑠆𑠇𑠈𑠉𑠊𑠋𑠌𑠍𑠎𑠏𑠐𑠑𑠒𑠓𑠔𑠕𑠖𑠗𑠘𑠙𑠚𑠛𑠜𑠝𑠞𑠟𑠠𑠡𑠢𑠣𑠤𑠥𑠦𑠧𑠨𑠩𑠪𑠫𑠬𑠭𑠮𑠯𑠰𑠱𑠲𑠳𑠴𑠵𑠶𑠷𑠸𑠺𑠹𑠻𑠼𑠽𑠾𑠿𑡀𑡁𑡂𑡃𑡄𑡅𑡆𑡇𑡈𑡉𑡊𑡋𑡌𑡍𑡎𑡏𑡐𑡑𑡒𑡓𑡔𑡕𑡖𑡗𑡘𑡙𑡚𑡛𑡜𑡝𑡞𑡟𑡠𑡡𑡢𑡣𑡤𑡥𑡦𑡧𑡨𑡩𑡪𑡫𑡬𑡭𑡮𑡯𑡰𑡱𑡲𑡳𑡴𑡵𑡶𑡷𑡸𑡹𑡺𑡻𑡼𑡽𑡾𑡿𑢀𑢁𑢂𑢃𑢄𑢅𑢆𑢇𑢈𑢉𑢊𑢋𑢌𑢍𑢎𑢏𑢐𑢑𑢒𑢓𑢔𑢕𑢖𑢗𑢘𑢙𑢚𑢛𑢜𑢝𑢞𑢟𑢠𑢡𑢢𑢣𑢤𑢥𑢦𑢧𑢨𑢩𑢪𑢫𑢬𑢭𑢮𑢯𑢰𑢱𑢲𑢳𑢴𑢵𑢶𑢷𑢸𑢹𑢺𑢻𑢼𑢽𑢾𑢿𑣀𑣁𑣂𑣃𑣄𑣅𑣆𑣇𑣈𑣉𑣊𑣋𑣌𑣍𑣎𑣏𑣐𑣑𑣒𑣓𑣔𑣕𑣖𑣗𑣘𑣙𑣚𑣛𑣜𑣝𑣞𑣟𑣠𑣡𑣢𑣣𑣤𑣥𑣦𑣧𑣨𑣩𑣪𑣫𑣬𑣭𑣮𑣯𑣰𑣱𑣲𑣳𑣴𑣵𑣶𑣷𑣸𑣹𑣺𑣻𑣼𑣽𑣾𑣿𑤀𑤁𑤂𑤃𑤄𑤅𑤆𑤇𑤈𑤉𑤊𑤋𑤌𑤍𑤎𑤏𑤐𑤑𑤒𑤓𑤔𑤕𑤖𑤗𑤘𑤙𑤚𑤛𑤜𑤝𑤞𑤟𑤠𑤡𑤢𑤣𑤤𑤥𑤦𑤧𑤨𑤩𑤪𑤫𑤬𑤭𑤮𑤯𑤰𑤱𑤲𑤳𑤴𑤵𑤶𑤷𑤸𑤹𑤺𑤻𑤼𑤽𑤾𑤿𑥀𑥁𑥂𑥃𑥄𑥅𑥆𑥇𑥈𑥉𑥊𑥋𑥌𑥍𑥎𑥏𑥐𑥑𑥒𑥓𑥔𑥕𑥖𑥗𑥘𑥙𑥚𑥛𑥜𑥝𑥞𑥟𑥠𑥡𑥢𑥣𑥤𑥥𑥦𑥧𑥨𑥩𑥪𑥫𑥬𑥭𑥮𑥯𑥰𑥱𑥲𑥳𑥴𑥵𑥶𑥷𑥸𑥹𑥺𑥻𑥼𑥽𑥾𑥿𑦀𑦁𑦂𑦃𑦄𑦅𑦆𑦇𑦈𑦉𑦊𑦋𑦌𑦍𑦎𑦏𑦐𑦑𑦒𑦓𑦔𑦕𑦖𑦗𑦘𑦙𑦚𑦛𑦜𑦝𑦞𑦟𑦠𑦡𑦢𑦣𑦤𑦥𑦦𑦧𑦨𑦩𑦪𑦫𑦬𑦭𑦮𑦯𑦰𑦱𑦲𑦳𑦴𑦵𑦶𑦷𑦸𑦹𑦺𑦻𑦼𑦽𑦾𑦿𑧀𑧁𑧂𑧃𑧄𑧅𑧆𑧇𑧈𑧉𑧊𑧋𑧌𑧍𑧎𑧏𑧐𑧑𑧒𑧓𑧔𑧕𑧖𑧗𑧘𑧙𑧚𑧛𑧜𑧝𑧞𑧟𑧠𑧡𑧢𑧣𑧤𑧥𑧦𑧧𑧨𑧩𑧪𑧫𑧬𑧭𑧮𑧯𑧰𑧱𑧲𑧳𑧴𑧵𑧶𑧷𑧸𑧹𑧺𑧻𑧼𑧽𑧾𑧿𑨀𑨁𑨂𑨃𑨄𑨅𑨆𑨇𑨈𑨉𑨊𑨋𑨌𑨍𑨎𑨏𑨐𑨑𑨒𑨓𑨔𑨕𑨖𑨗𑨘𑨙𑨚𑨛𑨜𑨝𑨞𑨟𑨠𑨡𑨢𑨣𑨤𑨥𑨦𑨧𑨨𑨩𑨪𑨫𑨬𑨭𑨮𑨯𑨰𑨱𑨲𑨳𑨴𑨵𑨶𑨷𑨸𑨹𑨺𑨻𑨼𑨽𑨾𑨿𑩀𑩁𑩂𑩃𑩄𑩅𑩆𑩇𑩈𑩉𑩊𑩋𑩌𑩍𑩎𑩏𑩐𑩑𑩒𑩓𑩔𑩕𑩖𑩗𑩘𑩙𑩚𑩛𑩜𑩝𑩞𑩟𑩠𑩡𑩢𑩣𑩤𑩥𑩦𑩧𑩨𑩩𑩪𑩫𑩬𑩭𑩮𑩯𑩰𑩱𑩲𑩳𑩴𑩵𑩶𑩷𑩸𑩹𑩺𑩻𑩼𑩽𑩾𑩿𑪀𑪁𑪂𑪃𑪄𑪅𑪆𑪇𑪈𑪉𑪊𑪋𑪌𑪍𑪎𑪏𑪐𑪑𑪒𑪓𑪔𑪕𑪖𑪗𑪘𑪙𑪚𑪛𑪜𑪝𑪞𑪟𑪠𑪡𑪢𑪣𑪤𑪥𑪦𑪧𑪨𑪩𑪪𑪫𑪬𑪭𑪮𑪯𑪰𑪱𑪲𑪳𑪴𑪵𑪶𑪷𑪸𑪹𑪺𑪻𑪼𑪽𑪾𑪿𑫀𑫁𑫂𑫃𑫄𑫅𑫆𑫇𑫈𑫉𑫊𑫋𑫌𑫍𑫎𑫏𑫐𑫑𑫒𑫓𑫔𑫕𑫖𑫗𑫘𑫙𑫚𑫛𑫜𑫝𑫞𑫟𑫠𑫡𑫢𑫣𑫤𑫥𑫦𑫧𑫨𑫩𑫪𑫫𑫬𑫭𑫮𑫯𑫰𑫱𑫲𑫳𑫴𑫵𑫶𑫷𑫸𑫹𑫺𑫻𑫼𑫽𑫾𑫿𑬀𑬁𑬂𑬃𑬄𑬅𑬆𑬇𑬈𑬉𑬊𑬋𑬌𑬍𑬎𑬏𑬐𑬑𑬒𑬓𑬔𑬕𑬖𑬗𑬘𑬙𑬚𑬛𑬜𑬝𑬞𑬟𑬠𑬡𑬢𑬣𑬤𑬥𑬦𑬧𑬨𑬩𑬪𑬫𑬬𑬭𑬮𑬯𑬰𑬱𑬲𑬳𑬴𑬵𑬶𑬷𑬸𑬹𑬺𑬻𑬼𑬽𑬾𑬿𑭀𑭁𑭂𑭃𑭄𑭅𑭆𑭇𑭈𑭉𑭊𑭋𑭌𑭍𑭎𑭏𑭐𑭑𑭒𑭓𑭔𑭕𑭖𑭗𑭘𑭙𑭚𑭛𑭜𑭝𑭞𑭟𑭠𑭡𑭢𑭣𑭤𑭥𑭦𑭧𑭨𑭩𑭪𑭫𑭬𑭭𑭮𑭯𑭰𑭱𑭲𑭳𑭴𑭵𑭶𑭷𑭸𑭹𑭺𑭻𑭼𑭽𑭾𑭿𑮀𑮁𑮂𑮃𑮄𑮅𑮆𑮇𑮈𑮉𑮊𑮋𑮌𑮍𑮎𑮏𑮐𑮑𑮒𑮓𑮔𑮕𑮖𑮗𑮘𑮙

The specimens in this section therefore use normal-width continuous text rather than isolated character strings. They exercise paragraph composition, line breaking, punctuation, regional glyph forms, Hangul composition, historical jamo, supplementary-plane ideographs, and mixed-script runs. A compact table at the end isolates shared Han codepoints for direct visual comparison.

6.1 Chinese

Mainland Simplified Chinese — Noto SC This paragraph uses the explicit Simplified Chinese family throughout.

现代中文排版并不只是把一个个汉字依次放进页面。一个真正的正文环境还需要同时处理标点、数字、书名号、引号、括号以及连续句子之间的换行关系，并且在较长的段落中保持稳定的字面和行距。对于共享的汉字码位，不同地区使用的字体还可能给出不同的字形，因此同一个“骨、直、令、具、角”在不同字体家族中会表现出细微而稳定的区域差异。这里使用一整段连续的简体中文来检查大陆简体字形、标点悬挂附近的布局、正常段落换行以及本地字体选择在长文本中是否能够始终保持有效，而不是只验证几个孤立字符能不能显示。

Mainland Traditional Chinese — Noto SC Traditional characters can also be deliberately typeset through the Mainland Chinese family. This separates character repertoire from regional glyph selection: traditional codepoints do not by themselves imply a Taiwan or Hong Kong font.

现代中文排版并不只是把一个个汉字依次放进页面。一个真正的正文环境还需要同时处理标点、数字、书名号、引号、括号以及连续句子之间的换行关系，并且在较长的段落中保持稳定的字面和行距。对于共享的汉字码位，不同地区使用的字体还可能给出不同的字形，因此同一个「骨、直、令、具、角」在不同字体家族中会表现出细微而稳定的区域差异。这里特意使用传统汉字而仍然选择大陆字体家族，以检查字符繁简与地域字形选择能否保持彼此独立，并观察长段落中共享汉字、繁体字和全角标点的连续排版。

Taiwan Traditional Chinese — Noto TC The Taiwan family is tested with continuous Taiwan Traditional Chinese rather than with a short regional-glyph sample.

正體中文的正文排版除了字符本身之外，也牽涉字型、標點、行距與段落節奏。當文件同時包含不同語言或不同地區的漢字時，程式不能僅依 Unicode 碼位猜測應採用哪一套地區字形，而應由明確的語言或字型選擇來決定。這個段落使用臺灣正體字型，

並刻意包含「骨、直、令、具、角、門、國、學、龍」等適合觀察地區字形差異的共享漢字，同時加入引號、頓號、逗號與句號，藉此確認較長的正體中文內容在正常欄寬中能自然換行，而且從段首到段末始終維持同一個本地 CJK family。

Document Traditional Chinese – Shanggu The following paragraph is deliberately left completely unwrapped. It therefore remains under the document CJK family, Shanggu, and provides the control against which the explicit Chinese, Japanese, and Korean families can be compared.

漢字進入現代數位排版之後，同一個編碼往往可以由許多不同的字體呈現，而字形的歷史層次、地域習慣和設計風格並不會因此消失。傳統中文尤其適合用來觀察這種情形：有些字在不同地區的正文中結構略異，有些字的筆形和部件位置不同，而另一些差別只有在一整段文字的灰度與節奏中才會顯現。這個段落不使用任何語言專用命令，使所有漢字直接落入文件的 Shanggu CJK family；因此它不只是檢查單字是否存在，也檢查文件級 CJK 選擇在較長的連續文本、標點和自動換行之間能否穩定維持。

6.2 Korean and Historical Hangul

Modern Korean – Hangul The first Korean specimen deliberately avoids Hanja. It tests ordinary modern Hangul composition, punctuation, word spacing, and paragraph wrapping.

한글로 쓰인 긴 문장은 여러 음절 블록이 끊임없이 이어지면서도 낱말 사이의 간격과 문장 부호의 위치가 자연스럽게 유지되어야 한다. 짧은 보기에서는 글꼴이 몇 글자를 제대로 보여 주는지만 확인할 수 있지만, 실제 문단에서는 줄 끝에서 낱말이 어떻게 배치되는지, 괄호와 따옴표가 앞뒤 글자와 어떻게 어울리는지, 여러 줄을 지나도 같은 글꼴 선택이 계속 유지되는지를 함께 살펴볼 수 있다. 이 문단은 완성형 한글이 이어지는 일반적인 한국어 본문을 충분히 길게 배치하여 문장 전체의 모양과 줄 간격을 확인하고, 뒤에 나오는 한자 혼용문과 옛한글 자료를 비교하기 위한 현대 한글 기준 표본으로 사용한다.

Modern Korean – mixed Hanja and Hangul The next paragraph keeps Hangul and Hanja inside a single explicit Korean family. Shared Han characters must therefore retain Korean regional forms without interrupting the surrounding Hangul text.

韓國語의 現代 文體는 오늘날 대체로 한글만으로 적지만, 歷史、言語、哲學、法律과 같은 분야에서는 漢字를 함께 적은 문헌을 어렵지 않게 만날 수 있다. 이런 漢字混用文을 조판할 때에는 한글 音節과 句讀點 사이에 들어가는 漢字가 갑자기 다른 地域 字形으로 바뀌지 않아야 하며, 韓國、朝鮮、漢字、文化、文字와 같은 어휘도 앞뒤의 한글 문맥과 한 family 안에서 자연스럽게 이어져야 한다. 이 문단은 國語의 읽기 흐름을 유지하면서 여

러 漢字語와 고유어를 섞어, 실제 문헌에서 볼 수 있는 긴 혼용문이 줄바꿈을 거처도 안정적으로 조판되는지를 시험한다.

Jeju language — corpus specimen Jeju is useful here because practical orthographies make active use of the arae-a and decomposed Hangul sequences that are rare in ordinary modern Korean. The following is a composite regression specimen assembled from short attested Jeju-language passages rather than a newly invented “standard” Jeju paragraph.

제주말 (濟州말) 은 제주섬서 글아지는 말이다. 제주말모음은 시간 지나명 여러 가지 변화가 이섯다. 제주섬은 한라산광 주깃디 오름덜록 일뤰진 섬이라. 옛적 화산활동으록 굴부리가 께썩 곤데곤데 으라 오름이 하영 샷다. 낭 (木) 은 나무질로 뉘 줄기를 가지고 잇인 여러해살이 식물이다. 제주말 위키백과 또시는 제주어 위키백과는 제주말로 운영험신 위키백과 언어판이라.

Middle Korean and Old Hangul — historical corpus specimen The historical specimen deliberately retains decomposed jamo and tone marks. Rather than normalizing several sources into modern spelling, short genuine historical fragments are placed consecutively to form a paragraph-level shaping regression.

나·랏·말·ㅅ·미·中·듀 。國·국·에 달·아문·쥬·와로서·르·스·뭇·디아·니·힐·썩 ·아기·빅·여·셔·과
굴·이·가·숨·알·파·주·글··뎡··허·거·든 두·들·젠 開·關·힐 켓·므·리 疏·通·호·고 남·근 古·今·엿 남·기·섯
것·도·다 머·리·알·포·고·고·히·막·고 머·리·와·눈·괘·환·티·아·니·커·든 ·그·운·이·우·호·로·울·아
머·리·알·포·미·뽻·리·는 ·뎡·거·든 넓·고·큰 天·造·스·이·에 다·슬·며 어·즈·러·우·민·들 엿·디·덜·덜·흔 數
| 리·오

The corpus paragraph above tests realistic historical material. The following synthetic boundary sequence has a different purpose: it explicitly crosses Hangul Jamo Extended-A, ordinary Jamo, and Extended-B while remaining inside one shaping run.

·맏

Modern decomposed control:

각

6.3 Japanese and Ryukyuan

Modern Japanese Modern Japanese combines kana, regional Han forms, punctuation, and frequent script changes inside the same sentence. The entire paragraph remains under the explicit Japanese family.

現代日本語の文章では、漢字、ひらがな、カタカナが一つの段落の中で何度も切り替わり、句読点や括弧、かぎ括弧も連続して現れる。短い文字列だけではフォントが表示できることしか分からないが、長い本文にすると、漢字と仮名の大きさの釣り合い、約物の位置、行末での折り返し、そして次の行へ移った後の字形の連続性まで確認できる。この段落では「日本語」「漢字」「文字」「学校」「文化」「骨」「直」「令」「具」「角」などを通常の文脈に入れ、日本語用の地域字形を保ったまま複数行の文章が安定して組版されるかを試している。

Historical kana orthography The opening sentence below is historical text; the continuation is a showcase-written regression passage that deliberately retains historical kana spellings such as ゐる, やう, and 言ふ. This keeps the specimen long enough for paragraph testing without presenting newly written material as an archival transcription.

真夏の宿屋は空虚であつた。窓の外には白い雲がゆつくり流れ、庭の木立では蟬が絶えず鳴いてゐる。旅人は古い椅子に腰をおろし、まだ來ない馬車の音を待ちながら、時計の針ばかりを眺めてゐた。誰も何も言はないので、午後の光だけが廊下を少しづつ移つて行くやうに見えた。かうして長い時間が過ぎても、玄關の戸は一度も開かず、遠くの道から聞えて來る物音だけが、時々静かな宿の空氣を動かしてゐるのであつた。

Japanese Kanbun The first sentence is taken from Rai San'yō's *Nihon Gaishi*; the remainder is a showcase-written Kanbun-style regression continuation. The purpose here is typographic: Japanese Han glyphs must remain selected even when no kana occur in the run.

外史氏曰、吾讀舊志、見鳥羽帝時數下制符。蓋日本文章之傳、漢籍之學與國史之記相因而久矣。昔之學者讀經史、作序記、論人物、述治亂、多用漢文而各成其體。今取其字列於一段、所以驗日本字形之連續、句讀之位置、長行之折返、以及共享漢字在無假名介入時亦不離其本地字體也。文字雖同出一碼、而骨直令具角門國學龍諸字之形、隨地域字體而微異、故尤宜以連續文章觀之。

Okinawan — corpus specimen Okinawan is represented by a composite specimen from the Wikimedia Okinawan test project. Its mixed use of kana and Han characters makes it a

useful additional Japanese-family regression, while the orthography should not be read as a claim that one universal Okinawan spelling standard exists.

沖縄口えー琉球諸語ぬてい壺ちやいびーん。うちなーぐちぬ首里那覇方言おー琉球国ぬじょーま言葉やびたん。琉球王國や、昔ジジ中心んかい存在さる王国やたん。王や首里天加那志とう言ん。首里城や、沖縄県ぬ城やん。琉球王朝ぬ王城っし、沖縄県ぬいっちゃん大さるぐしくやん。漢字えー、大昔唐ぬ国んかいそーじー現りーる文字ぬていーちやん。やまとう、高麗、るーちゅー、ベトナムおー、大昔唐ぬ国から漢字伝わていちゃん。

Ainu Ainu written in the modern katakana orthography makes extensive use of Katakana Phonetic Extensions for syllable-final consonants. The following specimen therefore exercises ordinary katakana together with characters such as ク, シ, ム, ラ, and the combining form プ inside continuous text.

イタクプリカ ソモネコロカ シサムモシリ モシリソカワ チヌムケニシパ
チヌムケ カッケマツ ウタペラリワ オカウシケタ クニネネワ アイヌイタク
アニ クイタクルウェ ネワネヤクン ラモッシワノ クヤイライケプ ネルウェ
タパンナ。エエパキタ カニアナクネ アイヌモシリ シシリムカ ニプタニコタ
ン コアパマカ クネルウェネ ウッチケクニプ ラカサッペ。

6.4 Vietnamese

The same constitutional passage is used throughout this subsection so that the specimens differ primarily in writing system or font selection rather than in linguistic content. Quốc Ngũ is first typeset with the document Latin family, Libertinus, and then with the dedicated Vietnamese family. The corresponding Hán-Nôm text uses the VinaWiki repertoire.

Quốc Ngũ – document Latin family The first specimen is deliberately unwrapped and therefore remains in the document Latin family, Libertinus. It tests Vietnamese diacritics without a language-specific font override.

Trải qua mấy nghìn năm lịch sử, Nhân dân Việt Nam lao động cần cù, sáng tạo, đấu tranh anh dũng để dựng nước và giữ nước, đã hun đúc nên truyền thống yêu nước, đoàn kết, nhân nghĩa, kiên cường, bất khuất và xây dựng nên nền văn hiến Việt Nam. Từ năm 1930, dưới sự lãnh đạo của Đảng Cộng sản Việt Nam do Chủ tịch Hồ Chí Minh sáng lập và rèn luyện, Nhân dân ta tiến hành cuộc đấu tranh lâu dài, đầy gian khổ, hy sinh vì độc lập, tự do của dân tộc, vì hạnh phúc của Nhân dân. Cách mạng tháng Tám thành công, ngày 2 tháng 9 năm 1945, Chủ

敵參，鋒靈縱窮邪戩戩，該鉅狎駁虬戩狎，髡髡諱風癢縱縱戩戩，癢諱穉穉隄隄
慨緝縱風戩。

6.6 WenJin Multi-Plane Coverage

WenJin is different from the language-specific families above. It is exposed as one public local family whose internal fallback configuration spans multiple CJK planes. The paragraph below is therefore intentionally a coverage stress test rather than a linguistic specimen: ordinary Han text is interleaved with supplementary-plane characters while the public command remains unchanged.

中國文字的編碼範圍遠遠超過基本多文種平面，因此只測試一二三天地玄黃這類常用字並不足以確認大型漢字字體的覆蓋是否正常。這一段先由普通漢字開始，隨後插入第二平面的亅、吉、土、堅、丕，再回到典籍文字與古今學術，之後繼續進入第三平面的丌、丰、𠂇、冂、𠂆。當不同平面的字符在同一個長段落中交錯出現時，使用者仍然只看見一個 \WJ family；Plane 0、Plane 2 和 Plane 3 的具體 fallback 應由內部配置完成，而不應要求正文作者為不同 Unicode plane 手動切換命令。

The following table complements the paragraph test with deterministic block coverage samples. For each CJK Unified Ideographs block, the first twenty encoded ideographs are shown. This makes missing coverage or an incorrect Plane 0, Plane 2, or Plane 3 fallback immediately visible and keeps the regression sample stable across releases.

6.7 Regional Forms of Shared Han Characters

The preceding specimens test normal paragraphs. Only here are shared codepoints deliberately isolated. Every cell below contains the same Unicode character; the visible difference should therefore come from the selected regional family rather than from different encoded variants.

7 Document Features

feature system (composable document modules) is demonstrated throughout this PDF, but this section gives each major public feature an explicit specimen as well. The showcase therefore tests not only isolated modules but also whether they coexist in a realistic document.

Table 5: WenJin CJK block coverage

Block	Range	Twenty-character specimen
CJK Unified Ideographs	U+4E00–U+9FFF	一丁丂七上丅丅万丈三上 下丅丅丅丅丅丅丅丅丅丅
Extension A	U+3400–U+4DBF	止因崑个牛×月𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔
Extension B	U+20000–U+2A6DF	𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔
Extension C	U+2A700–U+2B73F	𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔
Extension D	U+2B740–U+2B81F	𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔
Extension E	U+2B820–U+2CEAF	𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔
Extension F	U+2CEB0–U+2EBEF	𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔
Extension I	U+2EBF0–U+2EE5F	𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔
Extension G	U+30000–U+3134F	𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔
Extension H	U+31350–U+323AF	𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔
Extension J	U+323B0–U+3347F	𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔 𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔𠂔

Table 6: Regional forms of shared Han codepoints

Character	Shanggu	Noto SC	Noto TC	Japanese	Korean
骨	骨	骨	骨	骨	骨
直	直	直	直	直	直
令	令	令	令	令	令
具	具	具	具	具	具
角	角	角	角	角	角
門	門	門	門	門	門
國	國	國	國	國	國
學	學	學	學	學	學
龍	龍	龍	龍	龍	龍

7.1 Mathematics

Mathematical font selection is deliberately left at IMPE's default setting. The following single display combines several common mathematical alphabets, operators, delimiters, accents, matrices, integrals, sums, limits, radicals, and nested scripts as a compact stress specimen.

$$\begin{aligned}
\mathcal{Z}_{\beta,\mu} &:= \mathrm{Tr}_{\mathcal{H}} \exp \left[-\beta \left(\widehat{\mathbf{H}} - \mu \widehat{\mathbf{N}} \right) \right] \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \int_{\mathbb{R}^{3n}} \prod_{j=1}^n \frac{d^3 \mathbf{p}_j}{(2\pi\hbar)^3} \exp \left[-\sum_{j=1}^n \sqrt{m_j^2 c^4 + c^2 \|\mathbf{p}_j\|^2} \right] \\
&\quad \times \det \left(\begin{array}{ccc} \frac{\partial^2 \mathcal{L}}{\partial q^i \partial q^j} & \nabla_{\mu} F^{\mu\nu} & \langle \psi | \widehat{\mathbf{H}} | \phi \rangle \\ \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2} & \sqrt[3]{1 + \sqrt{1 + \lambda^2}} & \frac{\mathbb{E}[X^4] - 3\mathbb{E}[X^2]^2}{\mathrm{Var}(X)^2} \\ \oint_{|z|=1} \frac{\log(1+z)}{z^{r+1}} dz & \prod_{p \in \mathbb{P}} (1 - p^{-s})^{-1} & \lim_{\varepsilon \rightarrow 0^+} \frac{f(x+\varepsilon) - f(x-\varepsilon)}{2\varepsilon} \end{array} \right) \\
&\quad + \underbrace{\widetilde{\mathbf{A}}_{\mu}^{\dagger \nu} \overline{\mathbf{B}}_{\nu}^{\rho} \widehat{\mathbf{C}}_{\rho}^{\mu}}_{\text{accented tensor contraction}} + \frac{\Gamma(\frac{1}{2} + i\alpha) \zeta(2s)}{\sqrt{2\pi\sigma^2}} \\
&\quad + \begin{cases} e^{-1/x^2} \sin(\pi/x), & x \neq 0, \\ 0, & x = 0, \end{cases} \\
&\quad \xrightarrow[\alpha \rightarrow 0]{\beta \rightarrow \infty} \left(\bigoplus_{j=1}^r \mathbb{C}^{n_j} \right) \otimes \bigwedge^k V \\
&\quad \implies \mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\mathrm{Var}(X)}{t^2}, \quad t > 0, \\
&\quad \alpha, \beta, \gamma, \delta, \varepsilon, \theta, \vartheta, \lambda, \mu, \nu, \xi, \pi, \varpi, \rho, \varrho, \sigma, \phi, \varphi, \chi, \psi, \omega \in \mathbb{R}.
\end{aligned}$$

7.2 Tables

Table 7 uses the IMPE book-table environment rather than a raw `tabular` wrapper.

7.3 Lists and Example Blocks

A normal list remains ordinary $\LaTeX$:

1. automatic font ownership;

Table 7: Feature integration summary

Feature	Public route	Exercised here
Mathematics	<code>math</code>	yes
Tables	<code>tables</code>	yes
Images	<code>image</code>	yes
Citations	<code>citations</code>	yes
Index	<code>index</code>	yes
Hyperlinks	<code>hyperlinks</code>	yes
Running headers	<code>headers</code>	yes
List environments	<code>lists_envs</code>	yes

2. script-specific behavior where required;
3. reusable document features;
4. stable public interfaces.

The IMPE ExampleBlock environment provides a distinct indented specimen:

A canonical showcase should be readable as a document, useful as a visual regression specimen, and representative of the public interface that new documents are expected to use.

7.4 Images

The image feature is exercised with the standard $\text{\TeX}$ distribution’s example image when it is available. This keeps the canonical source self-contained without committing an otherwise meaningless binary illustration.

Figure 1 also exercises cross-reference hyperlinks.

7.5 Citations and Bibliography

Citation support is loaded through the feature system and configured before feature activation. A normal textual citation can therefore be written as Knuth (1984), while a parenthetical citation appears as (Knuth, 1984).

The bibliography is printed near the end of the document.

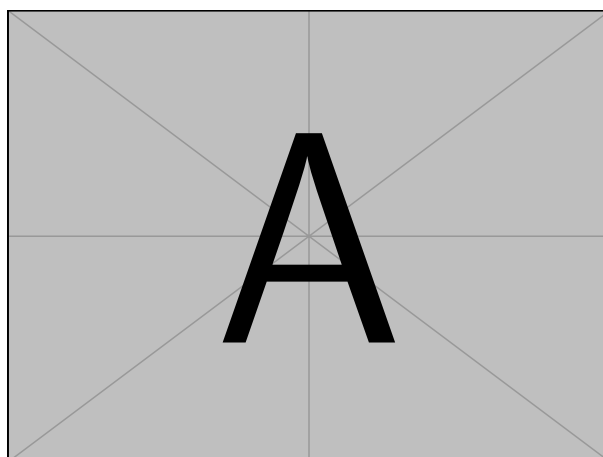


Figure 1: Image-feature specimen using the standard $\text{\TeX}$ example image.

7.6 Index Terms

Several concepts in earlier sections were inserted with the public `\Term` interface. For example, **canonical showcase** (release-time specimen) itself is now indexed. The generated index at the end of the PDF therefore also exercises indexed terms, optional descriptions, and hyperlinks back to their first occurrence.

7.7 Hyperlinks, Footnotes, and Running Headers

The table of contents is clickable, numbered headings carry stable targets, and section headings can link back toward their table-of-contents entries. The running header above this page is produced by the `headers` feature.

Footnote markers are also bidirectional.²

For an external-link specimen, the project repository can be represented with a normal hyperlink: [IMPE \$\text{\LaTeX}\$ System on GitHub](#).

References

Knuth, D. E. (1984). *The texbook*. Addison-Wesley.

²Clicking the footnote marker jumps to this note; the note marker can link back to the original position.

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